LapMDNet: Remote Sensing Object Detection Under Foggy Conditions via Physics-Guided Laplacian Prior and Masked Feature Distillation

Weizhi Yang, Jiangqun Ni, and Yi Xie

***Abstract*—For object detection in remote sensing images, foggy conditions tend to degrade image quality by scattering light and obscuring critical details, thereby compromising the performance of the involved detection models. To address this challenge, we first analyze the feature response of the Laplacian operation based on the atmospheric scattering model (ASM) and design two complementary Laplacian templates for bidirectional edge extraction and fuse them into a customized convolution kernel to enhance fog-degraded features. A rotated object detection method based on masked clean feature distillation is proposed, which leverages clean image features to facilitate the learning of fog-degraded image features. A dual-stream feature attention fusion module is then adopted to integrate the original yet blurred predistillation features with the clear but potentially noisy post-distillation features generated by the feature adjustment module, thus rendering the features for more effective object detection. Finally, a bidirectional attention-based feature pyramid network (BI-AFPN) is employed to enhance multilevel feature fusion. Extensive experiments on the dataset for object detection In optical remote-sensing images (DIOR)-Foggy, dataset for object detection in aerial images (DOTA)-Foggy, and real-world RDDTS fog datasets demonstrate that the proposed model outperforms other state-of-the-art methods.**

***Index Terms*—Feature distillation, foggy conditions, mask, object detection, physics-guided.**

I. Introduction

As ONE of the fundamental functions of intelligent systems, object detection has been widely applied in numerous fields, e.g., industrial defect detection [1], face detection, intelligent medical diagnosis, and autonomous driving [2], [3], [4]. Similarly, a growing body of research [5], [6], [7], [8] has applied object detection technology to the remote sensing field for urban governance, environmental monitoring, and uncrewed aerial vehicle (UAV) recognition, among other applications. Despite considerable advances in remote sensing object detection, its performance still degrades significantly under challenging weather conditions such as heavy fog. On the other hand, the backgrounds of remote sensing images are usually relatively complex, and the objects within them are dense. For example, there generally exist many small-sized targets, and their features are more prone to degradation under foggy conditions, thereby degrading detection performance.

Strategies designed for remote sensing object detection in fog-affected scenarios fall into three main categories: 1) directly applying conventional object detection models to remote sensing recognition; 2) performing joint inference of image dehazing models and object detection models in a pipeline manner; and 3) designing object detection models specifically for foggy conditions. The first method is the most straightforward, yet its detection performance is generally suboptimal. The second approach is intuitively feasible but has several drawbacks. First, the image dehazing process tends to cause degradation in target features, and the performance of the detector is contingent upon the efficacy of the image processing pipeline. Second, conducting joint inference in a pipeline paradigm leads to excessive structural complexity of the overall framework. Third, it is necessary to design different image processing model paradigms for different scenarios, which is unfeasible for practical applications. From the above analysis, the third approach seems to be a more promising one. Therefore, we propose a detection model tailored for remote sensing under foggy conditions, which is based on physics-guided Laplacian prior derivation and dual-stream feature fusion with masked feature distillation.

The main innovations of this work are summarized as follows.

1) We analyze the feature response of the Laplacian operation based on the atmospheric scattering model (ASM). The edge features extracted via Laplacian filtering from foggy images are close to those of haze-free images. Inspired by this observation, we design two complementary Laplacian templates for bidirectional edge extraction and fuse them into a customized convolution kernel. The constructed physics-guided Laplacian prior embeds imaging prior into feature learning and strengthens both positive and negative contour information of fog-degraded features.

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2) A feature distillation approach based on latent background masking is proposed. It leverages clean image features refined via the latent background feature mask to facilitate the feature learning of foggy remote sensing images, thereby capturing the image features that are more discriminative for the detection task.
3) A dual-stream feature attention fusion mechanism is employed to fuse the original yet blurred predistillation features and the clear but potentially noisy postdistillation features generated by the feature adjustment module, thus enabling the extracted features for detection to be more comprehensive and discriminative.
4) A bidirectional attention-based feature pyramid network (BI-AFPN) is put forward to adaptively integrate shallow detailed features and deep abstract features. This boosts the network's ability for detection of small objects within remote sensing images.
5) Extensive experiments are conducted on synthetic remote sensing datasets, including dataset for object detection In optical remote-sensing images (DIOR)-Foggy and dataset for object detection in aerial images (DOTA)-Foggy, as well as the real-world foggy dataset RDDTS. Experimental results demonstrate that the proposed method outperforms other state-of-the-art methods, and ablation studies further validate the effectiveness of each component in the proposed framework.

II. Related Works

A. Conventional Object Detection

As a pioneer in object detection models, the regions with convolutional neural network features (RCNN) series [9], [10], [11] adopts a two-stage process for detection tasks: first generating region proposals and then conducting refined detection. Representative detectors in the you only look once (YOLO) family, including YOLOv7 [12], YOLOv9 [13], and YOLOv10 [14], have emerged as widely adopted frameworks for object detection. They typically consist of a backbone network, a feature aggregation network, a detection head, and other related components. These models divide an image into grids, where each grid predicts multiple detection boxes and category probabilities. Zhang et al. [15] proposed LGA-YOLO, which incorporates a global context-aware module and a large-kernel local perception unit. Wang et al. [16] proposed the AUD-YOLO model for remote sensing image object detection by incorporating dynamic upsampling techniques.

However, in certain special scenarios such as those with densely distributed targets or where precise localization of targets is required, oriented bounding box (OBB) detectors are more suitable. Compared with generic object detection tasks, remote sensing or aerial image target detection is more challenging, mainly because objects in these images usually have arbitrary orientations and are densely distributed.

R3Det represents a classic oriented object detection model. Its feature refinement module (FRM) enables feature reconstruction and alignment via pixelwise feature interpolation. In addition, an approximate SkewIoU loss function is proposed, which solves the problem that SkewIoU calculation is nondifferentiable. Han et al. [17] propose a single-shot alignment network, which can generate high-quality anchors through an anchor refinement network and alleviate the inconsistency between classification score and localization accuracy by utilizing orientation-sensitive and orientation-invariant features. Xie et al. [18] proposed an oriented region proposal network (RPN) to generate oriented region proposals. Qian et al. [19] incorporated multiscale contextual information into the two-stage oriented object detection framework and developed a task-consistent focal loss, where the localization and recognition difficulty serves as the weight for the loss to boost the identification performance of hard samples. Zhou et al. [20] proposed an RSDB-Net, which acquires global information and local features through Swin Transformer and convolutional neural network branches, respectively, and fuses the features from these two branches. Xu et al. [21] and Yuan et al. [22] have conducted relevant research on small object detection and proposed corresponding solutions. Qian et al. [23] proposed three components, namely, the coarse-to-fine image-level supervised feature discriminative enhancement module, the discrete wavelet transform-based pointwise frequency attention module, and the coarse-to-fine instance-level supervised detection module, to refine object localization accuracy and reduce misclassification rates among visually similar subcategories. In addition, Qian et al. [24] proposed a transfer strategy to facilitate the transfer learning of bounding-box localization loss from horizontal object detection to oriented object detection.

These methods have achieved excellent detection performance in numerous conventional scenarios and fields. However, their detection performance is significantly compromised under foggy conditions.

B. Object Detection in Adverse Condition

To enable the detection model to maintain robust performance under adverse conditions, researchers have proposed relevant optimization schemes. Li et al. [25] proposed the AOD-Net model for image dehazing and built a joint optimization framework with the Faster R-CNN model, thereby improving detection performance. For foggy conditions, DS-Net [26] and TogetherNet [27] enhance detection performance under foggy conditions via joint learning of dehazing and detection models.

However, the integration of dehazing and detection tasks tends to lead to conflicts in parameter optimization because detection focuses on extracting target features from regions of interest rather than background features, whereas dehazing restoration requires extracting comprehensive, detailed features from the whole image. Consequently, such joint learning may slow down the convergence speed of the detection model and even degrade its detection performance.

Liu et al. [28] introduced IA-YOLO for object detection under adverse weather conditions, in which a lightweight convolutional module is employed to adaptively learn and regulate its parameters. Similarly, Kalwar et al. [29] proposed a GDIP module to enhance images to address adverse environmental conditions, such as fog. This module learns the parameters of multiple simultaneously executed image processing operations, obtains optimal weights to fuse their

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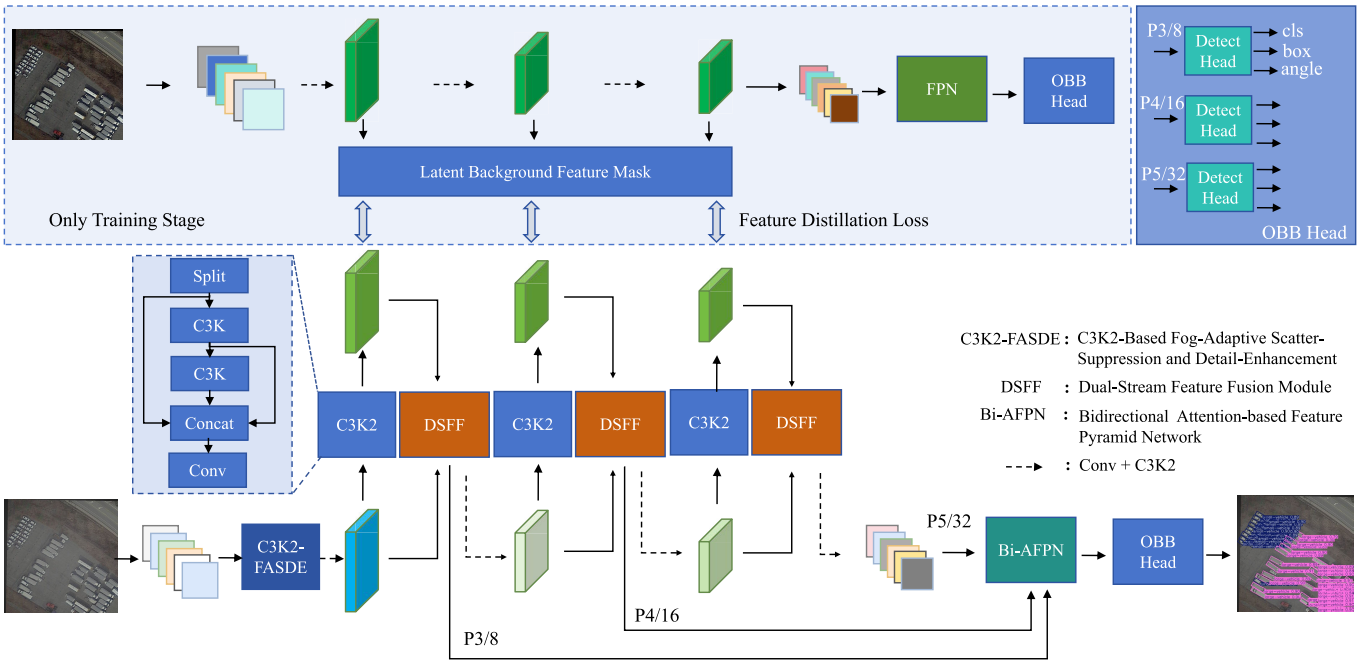


Fig. 1. Framework of our proposed method. The method comprises a clear-image detection network and a foggy-image detection network. Notably, the clear-image detection network is only used in the training stage and is excluded from the inference stage. We employ YOLOv11-OBB as the clear-image detection network, while the foggy-image detection network is improved based on YOLOv11-OBB, integrated with the proposed modules, including a C3K2 module that integrates the physics-guided Laplacian prior, the dual-stream feature attention fusion module, and the Bi-AFPN feature aggregation module.

outputs, and is further integrated into object detection networks. Li et al. [30] proposed the BAD-Net model, which consists of two feature branches. One branch first processes images through a defogging model and then extracts features from the defogged images, while the second branch performs direct feature extraction from fog-affected images. Dai et al. [31] proposed an underwater object detection method named GCC-Net, targeting low-light and turbid underwater scenarios. This method facilitates bidirectional feature communication and fusion between the enhanced image domain and the original domain. Furthermore, Zhang et al. [32] introduced the NIRNet model, which integrates a noise perception module and a dual-pathway feature fusion module. These two modules perform feature-weighted fusion according to the discrepancies between objects, respectively.

The aforementioned methods mainly improve object detection performance under harsh environments through image enhancement. However, there are two critical issues. First, the performance of the detection model depends on the effectiveness of the image processing model. For instance, in foggy conditions, the performance of image defogging and enhancement would gradually degrade as fog density increases. Second, image processing models tend to cause feature degradation, which is more evident in remote sensing images with more small objects. During image enhancement or restoration, feature noise is easily introduced, thereby compromising the performance of the involved object detectors.

III. Method

To achieve robust object detection under foggy conditions, we first adopt a physics-guided Laplacian prior to enhance reliable high-frequency features and then propose an oriented detection model based on feature distillation and dual-stream feature fusion. The proposed approach facilitates the learning of image features under foggy conditions and ensures the model's stability by incorporating postdistillation and predistillation features. The overview of the model is depicted in Fig. 1.

The model consists of a clear-image detection network and a foggy-image detection network. Notably, the clear-image detection network is exclusively deployed during the training phase and excluded from the inference phase. A C3K2 module that integrates the physics-guided Laplacian prior is introduced to process shallow-level features. Features extracted by the clear-image feature extraction network, following masking processing, facilitate the learning process of foggy-image features. The original yet blurred predistillation features and the clear but potentially noisy postdistillation features are fused via an attention mechanism to yield richer and more discriminative image representations. Within the feature aggregation network, an attention-driven bidirectional feature pyramid fusion strategy is adopted to adaptively integrate shallow-level detailed features and deep-level abstract features, as shown in Fig. 1.

A. Physics-Guided Laplacian Prior Derivation for Foggy-Image Enhancement

Foggy scenes suffer from significant degradation of target details due to the scattering effect of fog droplets on light, which follows the ASM, the fundamental physical model [33],

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[34] for foggy imaging, i.e.,

$$I(\mathbf{x}) = J(\mathbf{x}) \cdot t(\mathbf{x}) + A \cdot \big(1 - t(\mathbf{x})\big) \tag{1}$$

where $I(\mathbf{x}) \in \mathbb{R}^{H\times W\times 3}$ denotes the foggy image, $J(\mathbf{x}) \in \mathbb{R}^{H\times W\times 3}$ denotes the fog-free image, A denotes the global atmospheric light, and $t(\mathbf{x})$ denotes the transmission map characterizing fog concentration, which is determined by the scattering coefficient β of fog droplets and the scene depth $d(\mathbf{x})$ in accordance with the exponential attenuation law

$$t(\mathbf{x}) = e^{-\beta \cdot d(\mathbf{x})}. \tag{2}$$

A central difficulty in foggy-scenario detection lies in restoring the intrinsic high-frequency object details weakened by haze scattering. As a classical second-order differential operator, the Laplacian excels at capturing image edges and high-frequency details. Nevertheless, atmospheric scattering drastically attenuates high-frequency components under foggy conditions. It remains unclear whether high-frequency features can be reliably extracted directly from fog-degraded images. To answer this question, we conduct a theoretical analysis of the Laplacian operation based on the ASM.

To build the mathematical relation between Laplacian responses of fog-degraded and clear images, we apply the Laplacian operator ∇^2 to both sides of (1). Based on the product rule of the Laplacian operator ($\nabla^2(ab) = a\nabla^2 b + b\nabla^2 a + 2\nabla a \cdot \nabla b$), the left-hand side and the right-hand side of (1) can be expanded as follows:

$$\nabla^2 I(\mathbf{x}) = \nabla^2\Big(J(\mathbf{x}) \cdot t(\mathbf{x}) + A \cdot \big(1 - t(\mathbf{x})\big)\Big). \tag{3}$$

By expanding the product term using the Laplacian product rule, we obtain

$$\begin{aligned}\nabla^2 I(\mathbf{x}) = {} & J(\mathbf{x}) \cdot \nabla^2 t(\mathbf{x}) + t(\mathbf{x}) \cdot \nabla^2 J(\mathbf{x}) \\ & + 2\nabla J(\mathbf{x}) \cdot \nabla t(\mathbf{x}) + A \cdot \nabla^2\big(1 - t(\mathbf{x})\big).\end{aligned} \tag{4}$$

To facilitate mathematical derivation and follow common practice in haze-related studies, we adopt the locally constant assumption over a finite spatial neighborhood: both the atmospheric light A and transmittance $t(\mathbf{x})$ are slowly varying fields. Therefore, $\nabla t(\mathbf{x}) \approx 0$ and $\nabla^2 t(\mathbf{x}) \approx 0$. This approximation holds well for fog with gradual spatial variation, while approximation errors arise under nonuniform dense fog with sharp spatial changes of haze distribution. Although deviations arise from this idealized simplification, the subsequent deep feature learning module can learn robust representations from data and alleviate the degradation of detection performance caused by such approximation. Accordingly, (4) is simplified to

$$\nabla^2 I(\mathbf{x}) \approx t(\mathbf{x}) \cdot \nabla^2 J(\mathbf{x}). \tag{5}$$

Rearranging (5), we have the relationship between the Laplacian responses of fog-degraded and fog-free images

$$\nabla^2 J(\mathbf{x}) \approx \frac{\nabla^2 I(\mathbf{x})}{t(\mathbf{x})}. \tag{6}$$

Accordingly, the Laplacian high-frequency features of clear images can be calibrated from the Laplacian of hazy images with the transmission $t(\mathbf{x})$.

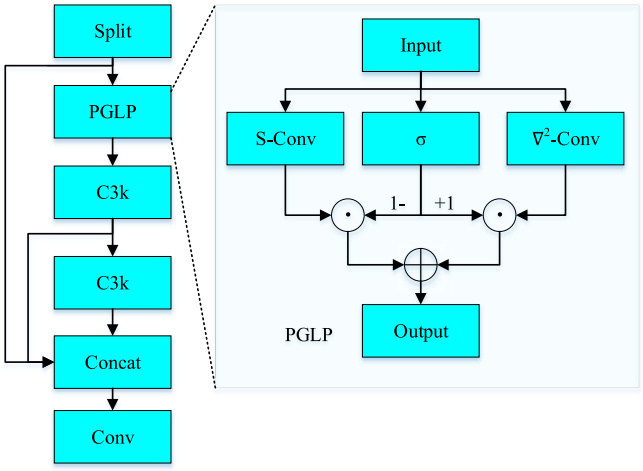


Fig. 2. C3K2-FASDE module.

Guided by this conclusion, we construct two dedicated Laplacian-based convolution kernels to impose a smoothness constraint on the fog background and a high-frequency representation on object details, respectively. A spatial adaptive gating mechanism is further embedded into the end-to-end network to integrate the above high-frequency calibration principle.

We then proceed to employ a discretized form of the Laplacian operator to facilitate model training. Specifically, the second-order central difference method is utilized, which approximates the second-order partial derivatives in the x- and y-directions using the differences between adjacent pixels. For the eight-neighborhood of the central pixel, the discrete Laplacian operator is derived as follows:

$$\nabla^2 = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}. \tag{7}$$

Complementary to the Laplacian detail enhancement kernel, we employ a scatter suppression kernel targeting the low-frequency fog scattering component in the ASM. The scatter suppression kernel is initialized as a center-negative 3×3 matrix

$$S = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}. \tag{8}$$

While the scatter suppression kernel shares structural similarities with the Laplacian kernel, their functional mechanisms are fundamentally different. The scatter suppression kernel is designed to suppress the low-frequency fog background ($A \cdot (1 - t(\mathbf{x}))$ in the ASM). In contrast, the Laplacian detail enhancement kernel targets the high-frequency target details ($J(\mathbf{x}) \cdot t(\mathbf{x})$).

Based on the C3K2 module in YOLOv11, along with the scatter suppression kernel and detail enhancement kernel, we propose a novel C3K2-based fog-adaptive scatter-suppression and detail-enhancement (C3K2-FASDE) module for foggy-image detection, as shown in Fig. 2.

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To achieve pixelwise adaptive fog suppression and detail enhancement, we design a fog-aware gating mechanism to modulate the outputs of the scatter suppression kernel and Laplacian detail enhancement kernel. Let $\phi_s(\mathbf{x})$ and $\phi_d(\mathbf{x})$ denote the pixelwise physical attention gates for scattering suppression and detail enhancement, respectively, where $\phi_s(\mathbf{x}), \phi_d(\mathbf{x}) \in [0, 1]$. A value close to 1 indicates dense fog regions that require strong suppression or enhancement, while a value near 0 represents fog-free regions that need almost no manipulation. The adaptive feature enhancement is formulated as

$$\mathbf{F}_{\text{enh}}(\mathbf{x}) = \left(1 - \phi_s(\mathbf{x})\right) \cdot \mathbf{F}_s(\mathbf{x}) + \left(1 + \phi_d(\mathbf{x})\right) \cdot \mathbf{F}_d(\mathbf{x}) \quad (9)$$

where $\mathbf{F}_s$ is the scattering-suppressed feature and $\mathbf{F}_d$ is the Laplacian-enhanced feature.

The Laplacian kernel with a central weight of 8 and surrounding coefficients of -1 contains large coefficient values. These large magnitudes excessively amplify edge features in convolution, which easily results in activation saturation and unstable gradient updates. Therefore, we multiply the entire kernel by a scaling factor of 0.01 to suppress the response amplitude. This normalization constrains the range of feature values and stabilizes network training. Furthermore, we combine a 90% Laplacian prior with 10% small-magnitude random noise. This configuration preserves the operator's core capability and introduces appropriate disturbance to facilitate model optimization and accommodate variations in different fog concentrations.

B. Feature Distillation Approach Based on Latent Background Feature Masking

The above derivation in the Laplacian domain establishes a high-frequency correlation prior between hazy images and clear images. However, such pixel-level differential constraints cannot be directly embedded into deep feature learning pipelines. To address this issue, this article introduces a feature distillation strategy: it refines and purifies the effective high-frequency edge information obtained from the Laplacian derivation, filters out redundant noise and false details caused by haze, and transfers the calibrated authentic feature details into deep semantic features via distillation, thereby achieving robust object detection under foggy conditions.

We first train a detector based on clear images. This detector then facilitates the backbone network in learning features from hazy images, enabling the network to infer clear features from blurred inputs. We perform feature distillation on hazy features at three different scales. The feature adaptation across these three scales operates in a cascaded manner: the large-scale clear features generated in earlier stages are propagated to subsequent small-scale feature maps, thereby achieving coarse-to-fine feature distillation.

Moreover, remote sensing images typically exhibit complex backgrounds. If distillation learning is performed on the entire feature map, the feature generation module will tend to focus on irrelevant background features, thereby distracting the model. To encourage the model to concentrate on dehazed feature extraction for valid targets and further improve detection performance, we adopt a latent background feature suppression distillation strategy. This method suppresses latent background features in the clear-image feature network to highlight potential target regions. The masked feature map then facilitates the distillation learning of the blurred feature map, ensuring that the distillation module focuses on target-related features. Specifically, a label map with a size of $H \times W \times 1$ is generated from the clear feature map via channelwise average pooling. Next, weakly activated feature points in this map are masked to 0, while others are retained as 1, yielding the final mask map. We perform an elementwise product between this mask map and the original feature map, yielding the latent background-suppressed feature map.

The acquisition process of the masked clear feature map is delineated as follows:

$$M(i, j) = \chi_\theta\left(\frac{1}{C}\sum_{c=1}^{C} F(i, j, c)\right) \quad (10)$$

where

$$\chi_\theta(x) = \begin{cases} 0, & x < \theta \\ 1, & else \end{cases} \quad (11)$$

$$F_m = F \odot M \quad (12)$$

where F, M, and F_m denote the original clear feature map, the mask map, and the masked clear feature map, respectively. Meanwhile, i, j, and c denote the horizontal position, the vertical position, and the channel position of feature points, respectively.

Given the significant discrepancy in feature domains between clear and foggy feature maps, directly aligning features would complicate and hinder the task; thus, we employ distribution alignment instead of domain alignment for feature distillation. Specifically, as shown in (13)–(15), for both clear and foggy feature maps, we subtract the spatialwise mean of each channel and then perform feature distillation learning

$$F_{md}(i, j, c) = F_m(i, j, c) - \overline{F_m(c)} \quad (13)$$

$$F_{sd}(i, j, c) = F_s(i, j, c) - \overline{F_s(c)} \quad (14)$$

where

$$\begin{cases} \overline{F_m(c)} = \dfrac{1}{H \times W}\displaystyle\sum_{i=1}^{H}\sum_{j=1}^{W} F_m(i, j, c) \\ \overline{F_s(c)} = \dfrac{1}{H \times W}\displaystyle\sum_{i=1}^{H}\sum_{j=1}^{W} F_s(i, j, c) \end{cases} \quad (15)$$

where $\overline{F_m(c)}$ and $\overline{F_s(c)}$ denote the mean values over channel c of the clear mask features and the features to be distilled, respectively.

We first train the YOLOv11-OBB model on the clear-image dataset for 300 epochs and save the well-trained weights. All parameters of this pretrained model are then fully frozen and serve as the teacher network. For subsequent training on hazy images, feature map distillation is adopted. Specifically, we perform distillation on feature maps at three different downsampling scales of the feature extraction network. Each foggy sample is matched with its corresponding original clear image, and the feature maps derived from paired clear images act as supervision to optimize the feature representation of

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hazy images. Note that this distillation module works only in the training stage and is removed during inference. In practical applications, only foggy images are fed into the model without any clear reference inputs.

C. Feature Distillation Loss Function

As illustrated in (16) and (17), the loss function for feature distillation is composed of the L1 loss and the Kullback–Leibler (KL) divergence loss. Specifically, the L1 loss is used to quantify the numerical discrepancy between defogged features and clear features, while the KL divergence is adopted to measure their distribution alignment. This integrated design empowers the model to grasp the global regularity of feature maps while preserving the precision of local values

$$L_{KL}(F_{md}, F_{sd}) = \sum_{c=1}^{C} \sum_{t=1}^{H\times W} \left[softmax\left(F_{md}[t,c]\right) \times \left(log\left(\frac{softmax(F_{md}[t,c])}{softmax(F_{sd}[t,c])} \right) \right) \right] \tag{16}$$

$$L_s = L1(F_{md}, F_{sd}) + L_{KL}(F_{md}, F_{sd}) \tag{17}$$

where F_{md} and F_{sd} denote the clean feature map and foggy feature map used for distillation, respectively.

D. Dual-Stream Feature Fusion Module

During the process of generating clear features from blurred features, valid features may be lost, or noise may be introduced, thereby degrading detection performance. Therefore, we adopt a dualstream feature attention fusion module (DSFAF) incorporating an attention fusion module to fuse the complete but blurred features before distillation and the clear but potentially noisy features after distillation, making the image features more informative and robust, as shown in Fig. 3. Dual-stream features are concatenated and fed into the attention module to refine channelwise attention weights. The attention module comprises two branches: salient feature attention based on global max pooling (GMP) and average feature attention based on global average pooling (GAP). Specifically, GMP effectively captures the key contours of targets, enhancing the ability to recognize small targets, while GAP preserves average features to strengthen global perception. Finally, the attention feature maps from both branches are aggregated to yield the output feature map.

E. Bidirectional Attention-Based Feature Pyramid Network

We conduct multilevel attention fusion on the shallow and deep features extracted by the backbone network, as shown in Fig. 4. First, the fused feature map aggregating features before and after distillation is fed into the BI-AFPN. The network leverages both top-down and bottom-up feature fusion strategies. In particular, multiscale feature maps are fused via a channel attention module rather than simple concatenation, allowing the model to adaptively learn the significance of each feature and further improve detection accuracy.

As illustrated above, our network follows a sequential processing pipeline. We first recover haze-blurred contour features via the customized Laplacian prior. Next, latent background masked distillation purifies high-level semantic representations. Then the dual-stream attention fusion achieves a favorable tradeoff between feature blur and noise interference, and the improved multiscale pyramid retains fine-grained details of small objects. Rather than mechanically stacking independent components, the entire framework forms an integrated and targeted solution for rotated object detection under foggy conditions.

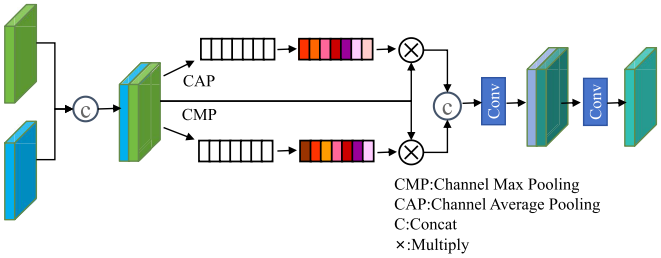


Fig. 3. Illustration of the proposed DSFAF module.

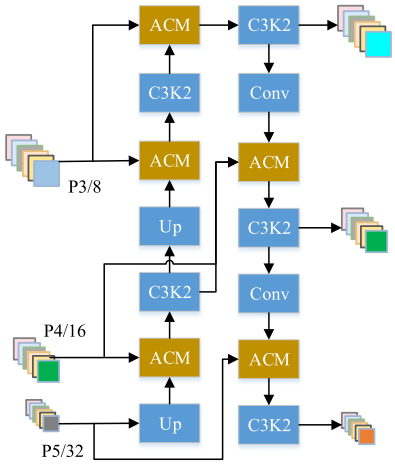


Fig. 4. Illustration of the proposed BI-AFPN network.

IV. Experiments

A. Datasets

Due to the challenges of collecting aerial or remote sensing images under foggy conditions and the lack of publicly available datasets, similar to [29] and [35], we synthesize foggy remote sensing object detection datasets based on the ASM [33], [34] and the original DIOR and DOTA datasets, which are denoted as DIOR-Foggy and DOTA-Foggy, respectively.

1) DIOR-Foggy: The DIOR dataset [36] comprises 20 object categories including expressway toll station (ETS), ship (SH), vehicle (VE), harbor (HA), golf field (GF), baseball field (BF), airplane (APL), airport (APO), windmill (WM) chimney (CH), storage (STO) tank, expressway service area (ESA), train station (TS), dam (DAM), overpass (OP), stadium (STA), basketball court (BC), tennis court (TC), bridge (BR), and ground track field (GTF). All images in the dataset have a uniform resolution of 800 × 800. The images are hazed using the ASM, as shown in Algorithm 1. Finally, 5862, 5863, and 11 738 pairs of clear–foggy images are allocated to the training, validation, and test sets, respectively. The dataset

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TABLE I

QUANTITATIVE COMPARISON OF DETECTION PERFORMANCE ON THE DIOR-FOGGY DATASET. THE BEST PERFORMANCE AND SUBOPTIMAL PERFORMANCE WERE HIGHLIGHTED IN BOLD OR UNDERLINED, RESPECTIVELY

Method	GTF	DAM	BR	APO	WM	BF	OP	SH	GF	CH	ETS	HA	ESA	TC	STA	BC	APL	VE	STO	TS	mAP
S2ANet	39.6	19.6	23.1	24.6	56.4	54.5	38.2	73.3	50.4	46.7	29.4	40.2	51.6	64.7	41.5	59.5	57.1	38.3	51.8	32.6	44.7
Sasm-Reppoints	44.2	21.6	21.5	26.0	64.3	60.7	36.4	75.0	53.2	49.5	32.8	43.4	56.1	70.2	40.5	61.9	60.7	42.8	55.6	34.8	47.6
YOLOv11-OBB	48.1	22.8	25.9	27.1	68.2	62.8	44.0	81.6	57.0	58.9	33.7	46.4	55.3	72.5	46.4	68.9	61.7	41.9	59.5	43.4	51.3
S2ANet + DCP	41.8	20.5	24.6	25.4	59.8	59.1	38.6	74.2	49.1	47.8	31.0	41.3	54.2	68.1	38.8	64.1	59.7	42.5	51.0	31.9	46.2
YOLOv11-OBB + FFA	49.4	23.8	26.8	**29.6**	68.9	65.2	42.8	80.1	57.7	62.2	32.6	46.3	57.5	70.4	46.0	69.2	62.9	41.4	60.9	43.0	51.8
YOLOv11-OBB + DCP	51.0	21.9	27.2	27.9	70.2	66.1	45.3	83.3	55.4	61.1	35.1	47.7	59.1	75.3	42.8	69.7	62.5	44.3	60.3	39.1	52.3
TogetherNet-OBB	51.0	24.3	26.1	26.7	70.1	63.3	45.0	81.5	58.2	62.3	34.4	46.8	55.3	72.4	43.8	69.3	65.6	42.8	62.5	41.2	52.1
GDIP-OBB	45.4	20.6	24.8	26.3	64.5	63.1	42.8	80.2	55.7	58.2	31.5	45.9	54.4	70.0	42.1	63.4	60.8	42.6	60.1	41.3	49.7
ACLNet-OBB	49.6	23.7	27.2	26.3	70.0	65.4	44.7	80.3	57.9	58.6	35.6	48.5	54.8	72.1	45.3	67.2	64.2	43.4	60.0	42.5	51.9
NIRNet	49.7	21.5	27.8	28.3	71.5	62.8	44.6	81.9	59.1	62.5	35.1	47.2	55.8	73.7	45.7	69.0	66.2	41.7	62.5	40.6	52.4
MASFNet-OBB	50.3	22.6	29.4	27.8	**71.9**	65.7	45.8	82.3	58.5	58.7	36.6	49.1	56.3	71.9	46.8	68.2	65.0	45.6	61.2	40.3	52.7
LapMDNet (Ours)	**51.4**	**27.0**	**31.7**	29.0	71.6	**66.9**	**46.4**	**83.4**	**61.3**	**64.1**	**37.7**	**50.5**	**59.9**	**76.2**	**50.5**	**73.1**	**67.4**	**47.6**	**66.1**	**45.4**	**55.4**

Algorithm 1 Synthesis of Foggy Remote Sensing Images Based on ASM

Input: J: Clean remote sensing image, A: Global atmospheric light, β: Atmospheric scattering coefficient.
Output: I: Synthesized foggy remote sensing image
$(H, W, C) = J.shape$
$s = \sqrt{\max(H, W)}$
$A = range(0.4, 0.7)$
$\beta = range(0.12, 0.2)$
for i in range(H) **do**
  for j in range(W) **do**
    $d = -0.04 \times \sqrt{\left(i - \frac{H}{2}\right)^2 + \left(j - \frac{W}{2}\right)^2} + s$
    $t = e^{-\beta \times d}$
    $I[i][j] = J[i][j] \times t + A \times (1 - t)$
  end for
end for

partition strategy remains consistent with that of the original DIOR dataset.

2) DOTA-Foggy: The DOTA dataset [37] comprises 15 object categories including GTF, BC, TC, storage tank (ST), soccer ball field (SBF), small vehicle (SV), large vehicle (LV), plane (PL), roundabout (RA), HA, SH, swimming pool (SP), BR, baseball diamond (BD), and helicopter (HC). The images are first cropped to a resolution of 640×640 and then hazed using the atmospheric scattering model. Considering the training efficiency, we only select images with more than five objects for training to accelerate model convergence. Finally, 5401, 8190, and 1556 paired clean–foggy images are allocated to the training, validation, and test sets, respectively.

3) HazyDet: The HazyDet dataset [38] contains 11 000 pairs of clear and foggy images, together with RDDTS—a subset consisting of 600 UAV-captured images taken under real foggy weather. The dataset covers three object categories: car, truck, and bus, with horizontal bounding boxes as annotations. All images are cropped to a uniform resolution of 600×600 pixels for the experiments.

B. Implementation Details

We conducted separate training and validation on each dataset to evaluate detection performance. In addition, we conducted cross-dataset generalization experiments by training on the DIOR-Foggy dataset and directly evaluating on the DOTA-Foggy dataset with no fine-tuning. Owing to the differences in category composition between the DIOR-Foggy and DOTA-Foggy datasets, we perform cross-dataset validation on a shared subset of representative common categories, including GTF, BR, SH, HA, TC, BC, PL, LV/SV, and ST. Furthermore, to verify the generalization capability and practical performance of our model under real foggy conditions, we train the model on partial synthetic fog datasets from HazyDet and conduct validation on real foggy images of the RDDTS subset. The number of training epochs is set to 300, the batch size to 8, and the early stopping patience to 50 epochs. The model is optimized by stochastic gradient descent (SGD), with an initial learning rate and decay factor both set to 0.01, momentum of 0.937, and three warm-up epochs.

C. Comparison With Other Methods

To begin with, we validate the effectiveness of our approach by conducting comparative experiments on the DIOR-Foggy dataset with several representative oriented object detection methods. The involving competing methods are categorized into three types, as detailed in Table I: 1) methods for direct detection in foggy scenarios, including S2ANet, Sasm-Reppoints [39], and YOLOv11-OBB; 2) "Defogging + Detection" frameworks: these methods first perform defogging via dark channel prior (DCP) [40] or FFANet [41] and then conduct object detection; and 3) integrated defogging and detection networks, such as TogetherNet, GDIP, ACLNet-OBB [42], NIRNet [32], and MASFNet-OBB [43].

Among methods that directly employ oriented detectors, YOLOv11-OBB achieves the most competitive performance, with a mean Average Precision (mAP) of 51.3. This demonstrates that the advanced backbone design of YOLOv11-OBB enhances its ability to extract discriminative features from foggy images.

The combination of an image defogging model and a target detection model is employed, which improves the detection performance on the basis of the original model to a certain extent, yet the improvement is marginal. This marginal improvement is attributed to two factors: first, the detection performance is highly dependent on the defogging effect, and defogging methods struggle to fully restore the details of targets under heavy fog; second, the independent

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Fig. 5. Qualitative comparison of detection performance on the DIOR-Foggy dataset. From left to right, the detection results correspond to YOLOv11-OBB, DCP + YOLOv11-OBB, NIRNet, MASFNet-OBB, ours, and ground truth, respectively.

defogging process often leads to the degradation of target-related features, resulting in missed detections of small or low-contrast targets.

Similarly, the joint training approach of defogging and target detection also offers certain advantages over the direct detection method, yet the gains remain modest. The fundamental reason for the unsatisfactory performance of joint methods lies in the inherent conflict between defogging and detection tasks. Defogging tasks optimize restoration loss, which requires extracting comprehensive detail features (including background) to improve image visual quality; in contrast, detection tasks rely on classification loss and regression loss, which demand prioritizing target region features while suppressing background interference.

In contrast, our method achieves an mAP of 55.4, outperforming all other comparative methods. This superiority is mainly attributed to utilizing physically driven Laplacian priors to optimize the feature learning of the module and the introduction of a clear feature generation task integrated into the target detection framework. Specifically, the clear feature generation task and target detection are two synergistic subtasks rather than conflicting ones. Furthermore, we implement effective fusion based on a dual-stream feature attention mechanism.

We conduct intuitive visualization and qualitative analysis on the detection results of different models on the DIOR-Foggy dataset. Fig. 5 illustrates the detection performance comparison under four distinct fog intensity levels: the first group denotes light fog, the second and third groups represent moderate fog, and the fourth group corresponds to heavy fog. From the visualization results, in the first comparison group, our approach successfully detects nearly all aircraft instances, whereas other competing methods suffer from numerous missed detections of airplane (APL) targets. In the second group of comparisons, our method only misses three extremely small STO tank targets, achieving superior detection performance compared to other methods. Similarly, in the third group of comparisons, our method successfully detects all targets, whereas other methods exhibit varying degrees of missed detections and even several false detections. In the fourth group under heavy fog conditions, the other three methods all suffer from severe missed detections for vehicle (VE) and TC targets. In contrast, our method accurately detects all three TC targets and identifies the majority of VE targets.

Fig. 6 provides an intuitive visualization of the feature heatmaps generated by our method and the comparative methods. As illustrated in the figure, under foggy conditions, backgrounds or nontarget objects introduce significant

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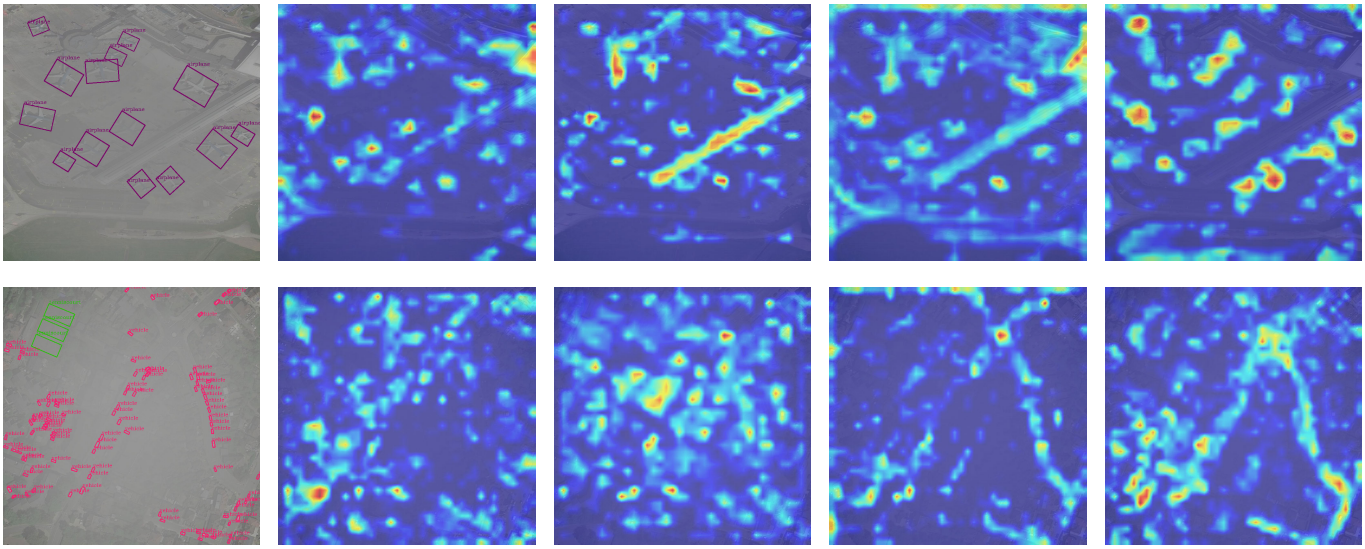


Fig. 6. Ground truth and heatmap of YOLOv11-OBB, NIRNet, MASFNet-OBB, and our method, respectively.

TABLE II

QUANTITATIVE COMPARISON OF DETECTION PERFORMANCE ON THE DOTA-FOGGY DATASET. THE BEST PERFORMANCE AND SUBOPTIMAL PERFORMANCE WERE HIGHLIGHTED IN BOLD OR UNDERLINED, RESPECTIVELY

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
S2ANet	79.6	43.5	2.8	21.8	66.3	85.6	86.5	79.5	20.2	59.6	24.8	21.9	64.9	39.7	29.2	48.4
YOLOv11-OBB	92.2	74.3	9.4	44.3	78.2	90.4	93.8	95.4	58.0	82.4	43.3	28.1	86.4	56.9	26.8	64.0
S2ANet + DCP	81.2	45.8	3.2	22.6	68.4	89.2	85.9	83.2	24.3	61.7	25.6	23.4	66.1	41.5	31.1	50.2
YOLOv11-OBB + FFA	90.9	73.5	12.6	35.3	78.8	90.9	94.7	95.8	61.4	82.1	43.0	30.7	85.8	**62.0**	25.0	64.2
YOLOv11-OBB + DCP	**92.7**	76.4	16.4	32.6	78.6	90.6	95.9	95.6	58.5	81.9	43.3	39.3	84.7	56.6	**44.3**	65.8
TogetherNet-OBB	92.4	**79.0**	19.1	42.7	**80.5**	90.7	95.8	95.3	61.6	83.4	37.9	32.3	84.8	49.3	32.0	65.1
GDIP-OBB	90.2	70.6	8.1	37.5	77.3	89.3	90.6	90.8	55.2	80.7	46.8	29.7	85.1	50.8	28.5	62.1
NIRNet	90.5	76.8	9.7	41.5	78.4	90.7	91.2	93.4	57.4	84.1	46.8	25.9	87.2	52.7	31.6	63.9
MASFNet-OBB	90.4	77.2	10.3	43.1	77.9	90.2	93.0	94.3	58.1	83.8	47.1	30.7	**87.6**	51.9	36.7	64.8
LapMDNet (Ours)	91.6	78.0	**19.7**	**49.0**	79.0	**91.5**	**96.4**	**95.9**	**65.5**	**84.2**	**50.3**	**41.6**	87.0	54.1	41.3	**68.3**

TABLE III

QUANTITATIVE COMPARISON OF DETECTION PERFORMANCE. THE MODELS ARE TRAINED ON THE DIOR-FOGGY DATASET AND DIRECTLY APPLIED TO THE DOTA-FOGGY DATASET

Method	GTF	BR	SH	HA	TC	BC	PL	LV/SV	ST	mAP
YOLOv11-OBB	28.2	3.1	83.1	39.3	86.4	**46.8**	73.4	40.6	57.8	51.0
S2ANet + DCP	25.7	2.6	80.4	35.7	83.8	43.2	71.3	42.5	55.2	48.9
YOLOv11-OBB + DCP	**39.9**	5.4	86.1	40.2	90.1	46.0	76.2	46.9	**64.0**	55.0
TogetherNet-OBB	31.5	3.8	83.5	39.0	85.2	46.7	74.6	47.5	60.9	52.5
NIRNet	28.4	5.3	83.8	40.1	88.3	45.9	76.3	47.3	60.4	52.9
MASFNet-OBB	33.6	4.9	84.3	41.6	89.7	44.8	76.0	46.4	59.7	53.4
LapMDNet (Ours)	39.7	**5.6**	**86.6**	**43.5**	**91.3**	45.5	**76.5**	**48.0**	61.2	**55.3**

interference to the comparative methods. This interference diverts the attention of the models and results in ambiguous feature activation in certain target regions, ultimately misguiding the detection process and causing false positives (FPs) or false negatives. In contrast, our method more effectively focuses on feature activation within target regions. Even for small targets such as VEs, it maintains robust detection performance.

To further validate the generalization ability of our method on different datasets, additional experiments are conducted on the DOTA-Foggy dataset, with corresponding comparison results summarized in Table II. It can be observed that our approach still attains the highest detection accuracy among all competing methods, yielding an mAP of 68.3.

As shown in Fig. 7, we also conduct intuitive visualization and qualitative analysis on the detection results of different models on the DOTA-Foggy dataset. In the first comparison group, YOLOv11-OBB failed to detect the ground-track-field targets. Although NIRNet and YOLOv1-1OBB + DCP successfully detected these targets, their detected positions and orientations exhibited considerable deviations from the ground truth. In contrast, our model not only successfully detected the targets but also achieved a high IoU between its detected positions and orientations and the ground truth.

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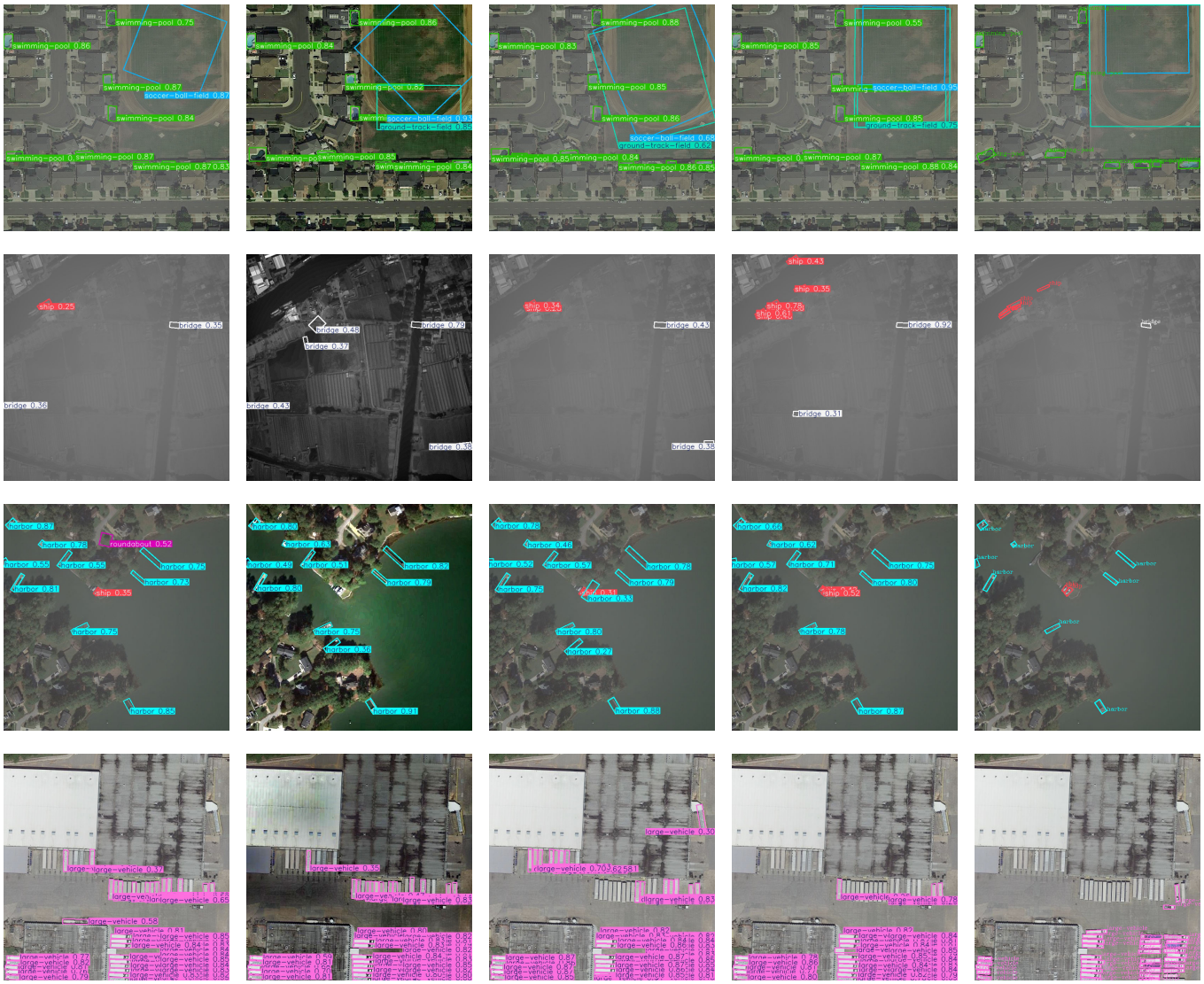


Fig. 7. Qualitative comparison of detection performance on the DOTA-Foggy dataset. From left to right, the detection results correspond to YOLOv11-OBB, DCP + YOLOv11-OBB, NIRnet, ours, and ground truth, respectively.

In the second and third comparison groups, despite the interference of fog, our model still accurately detected multiple SH targets, whereas the other comparative models exhibited missed detections. In the fourth comparison group, due to the small target size and the presence of interfering objects with morphologies similar to those of LV targets, the comparative models produced a significant number of FPs. Conversely, our model only produced a minimal number of FPs. Overall, these results demonstrate the superiority of our method.

To assess the generalization of the proposed approach, we directly tested the model trained on the DIOR-Foggy dataset on the DOTA-Foggy dataset without additional training. As shown in Table III, the results demonstrate that our model still achieves better detection performance.

To further evaluate generalization capability on real-world haze scenes, we conduct validation on the real fog dataset RDDTS, with quantitative results shown in Table IV. It is observed that the proposed model achieves better detection performance against other competing methods on authentic foggy samples, further verifying the generalization capability of our method.

TABLE IV
QUANTITATIVE COMPARISON OF DETECTION PERFORMANCE ON THE RDDTS DATASET

Method	Car	Truck	Bus	mAP
YOLOv11	72.0	17.5	22.5	37.3
YOLOv11 + DCP	**75.5**	19.6	25.3	40.1
GDIP	68.2	13.3	19.4	33.6
ACL-Net	72.4	15.9	24.8	37.7
MASFNet	73.4	<u>20.1</u>	<u>28.0</u>	<u>40.5</u>
LapMDNet-HBB (Ours)	<u>75.0</u>	**22.6**	**30.8**	**42.8**

D. Ablation Experiment

1) Effectiveness of Each Component: We adopted YOLOv11-OBB as the baseline and conducted a series of ablation experiments to verify the specific effectiveness of

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TABLE V
ABLATION EXPERIMENT RESULTS ON THE DIOR-FOGGY AND DOTA-FOGGY DATASETS

PGLP	Masked Feature Distillation	BI-AFPN	DSFAF Module	DIOR-Foggy	DOTA-Foggy	Params(M)	FPS
-	-	-	-	51.3	64.0	20.9	74
✓	-	-	-	52.9	65.8	21.0	71
-	✓	-	-	53.3	65.9	25.8	64
-	-	✓	-	51.9	64.8	22.5	69
-	✓	✓	-	53.8	66.5	27.4	60
-	✓	-	✓	54.2	66.7	27.8	58
✓	✓	✓	✓	55.4	68.3	29.7	51

TABLE VI
ABLATION RESULTS OF THE MASKING OPERATION ON THE DOTA-FOGGY DATASET

Feature Distillation	Mask Operation	mAP ↑
-	-	64.0
✓	-	64.8
✓	✓	65.9

TABLE VII
ABLATION RESULTS OF THE COMPLEMENTARY LAPLACIAN TEMPLATES ON THE DIOR-FOGGY DATASET

Baseline	Standard Laplacian	Complementary Laplacian Templates
51.3	52.1	52.9

TABLE VIII
ABLATION STUDY OF PROPOSED MODULES WITH VARIOUS FOG DENSITIES ON DIOR-FOGGY

Method	Light	Moderate	Heavy
YOLOv11-OBB	64.9	63.4	59.4
+PGLP	65.8	64.7	60.6
+MFD	67.6	66.5	62.5
+BIAFPN	65.2	64.3	59.8
LapMDNet	68.6	67.3	63.7

TABLE IX
ABLATION STUDY OF THE PGLP MODULE ACROSS VARIOUS MODELS ON THE DIOR-FOGGY DATASET

Method	Bseline	+ PGLP
YOLOv11m-OBB	51.3	52.9
YOLOv8m-OBB	51.4	52.7

the proposed method. We first validated each component individually and then performed validation through combined components, with the results summarized in Table V. When individually incorporating physics-guided laplacian prior (PGLP), masked feature distillation (MFD), and BI-AFPN on the DIOR-Foggy dataset, the mAP increased from 51.3 to 52.9, 53.3, and 51.9, respectively. When individually incorporating PGLP, masked feature distillation, and BI-AFPN on the DOTA-Foggy dataset, the mAP increased from 64.0 to 65.8, 65.9, and 64.8, respectively. By integrating different components, the mAP is further boosted, with the final model achieving 55.4 on DIOR-Foggy and 68.3 on DOTA-Foggy, respectively. Compared with the YOLOv11-OBB baseline with 20.9 M parameters, the model embedded solely with the proposed PGLP kernel module adds only 0.1 M parameters to reach 21.0 M. The variant equipped merely with the BI-AFPN feature fusion structure totals 22.5 M parameters, a mild increment of 1.6 M. Both modules incur only minor computational overhead and enhance fog-related feature extraction. After integrating all four proposed modules together, the final model has a total of 29.7 M parameters. This moderate parameter growth leads to substantial detection gains under heavy fog, achieving a favorable tradeoff between detection accuracy and inference efficiency.

Moreover, ablation experiments are conducted to evaluate the effectiveness of the clear feature masking module in the distillation procedure, as detailed in Table VI. Compared with the baseline, merely adding the feature distillation operation improved the mAP from 64.0 to 64.8; after incorporating the latent background feature masking, the mAP was further boosted to 65.9.

To further analyze the effectiveness of the proposed physically inspired Laplacian templates, we conduct comparative experiments with the standard fixed Laplacian convolution kernel under identical training configurations and parameter budgets. As listed in Table VII, our complementary Laplacian templates can achieve higher mAP compared with the standard fixed Laplacian kernel.

To verify the model's robustness under varying haze severity, we synthesize three independent subsets of 2000 images corresponding to light, moderate, and heavy fog by adjusting the parameters of the transmission field $t(\mathbf{x})$ within the ASM for evaluation. Table VIII shows mAP performance of the YOLOv11-OBB baseline, variants with single PGLP, MFD, BI-AFPN, and the full model integrating all modules. Among the three core components, MFD achieves the most prominent performance improvement across fog levels, followed by PGLP, while BI-AFPN brings moderate incremental gains. The full model accumulates the advantages of all modules, and its performance superiority over the baseline is further amplified under extreme heavy fog, which verifies the strong fog robustness of the proposed method.

2) Model-Level Ablation Comparison: We also applied the proposed components to different baseline models on the DIOR-Foggy or DOTA-Foggy datasets. The corresponding experimental results are presented in Tables IX–XI. When

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TABLE X
ABLATION RESULTS OF THE MFD AND DSFAF MODULES ON THE DOTA-FOGGY DATASET

Method	Bseline	+ MFD + DSFAF
YOLOv8s-OBB	61.9	64.5
YOLOv5s-OBB	60.2	63.1

TABLE XI
ABLATION RESULTS OF THE BIAFPN ON THE DOTA-FOGGY DATASET

Method	Bseline	+ BIAFPN
YOLOv8s-OBB	61.9	63.4
YOLOv5s-OBB	60.2	62.1

the PGLP was applied to YOLOv11m-OBB and YOLOv8m-OBB, the mAP was improved by 1.6% and 1.3% points, respectively. When the MFD and DSFAF components were applied to YOLOv8s-OBB and YOLOv5s-OBB, the mAP was improved by 2.6% and 2.9% points, respectively. In contrast, the integration of the BIAFPN module led to mAP improvements of 1.5% and 1.9% points, respectively.

V. CONCLUSION

In this article, we propose an object detection method based on a physics-guided Laplacian prior and masked clear feature distillation. Starting from the ASM, we construct two complementary Laplacian kernels for bidirectional edge enhancement and fuse them into a customized convolution module to strengthen haze-degraded contour features. The method leverages clear features to facilitate the learning of foggy-image features. A dual-stream feature attention fusion module is adopted to integrate the complete yet blurred predistillation features with the clear but potentially noisy postdistillation features generated by the feature adjustment module, thus rendering the image features more suitable for effective object detection. In addition, a BI-AFPN is adopted to enhance multilevel feature fusion. Comprehensive evaluations on the DIOR-Foggy, DOTA-Foggy, and real-world RDDTS datasets validate the superiority of our model over existing state-of-the-art approaches.

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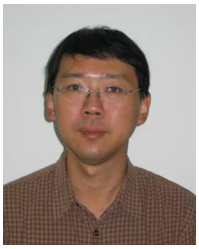
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